

Matthew Choi

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EXPERIENCE

Meta

Software Engineer, Machine Learning

Aug 2024 - Mar 2026

- **Built and shipped an ML classifier for detecting compromised high-profile Facebook Pages**, recovering **10x** more pages and doubling precision (**31% → 70%**) while achieving daily detection over the previous strategy's cadence of 3 days.
- **Prototyped an LLM-based bulk inference pipeline** that cut scaled reviewer workload by **30%** at **99%** precision for page compromise cases, informing the team's production review strategy.
- **Led the end-to-end redesign of a system that proactively queues content-creator impersonator accounts for takedown**, combining ML signals and heuristics into a unified priority-ranking model.
 - Increased action rate (precision) **2.4x**, from **14.5%** to **34.5%**.
 - Improved recall **6.7** times from **14%** to **93.6%** by sourcing new impersonator-candidate signals, reducing victim prevalence by **12%** of the team-wide goal of **30%**.
- **Partnered with cross-functional ML infra teams and prompt engineers** to enable multi-media inputs for Llama 4 bulk inference, build shared datasets and LLM evaluation tooling for rapid prompt iteration, and automate ablation studies.

Meta

Software Engineer Intern

May 2022 - Aug 2022

- **Reduced demographic bias in a face-verification deep learning model** for Meta Reality Labs
- **Researched and implemented state of the art loss functions and convolutional neural network architectures**, and increased performance from **95.0%** to **98.2%** with a model **five** times smaller than the one previously used.
- **Built an internal analysis toolbox** using PyTorch Lightning, TensorBoard, and Matplotlib to automate bias-metric visualizations and facial-embedding interpretability (Grad-CAM, t-SNE, PCA, LDA).

Birch.ai

AI Researcher Intern

Sept 2021 - Nov 2021

- **Built a synthetic data pipeline** that generated documents with tabular content and extracted structured data for model training.
- **Trained computer vision models** using the PyTorch-based MMDetection toolbox to detect tables and signature blocks in healthcare documents.

Intuit

Software Engineer Intern;

May 2021 - Aug 2021

- **Contributed to Intuit's tax knowledge engine** by applying calculation and completeness graphs for knowledge engineering.
- **Built a version-control system for knowledge-engineering artifacts** using React, Node.js, and Express with GitHub integration.
- **Implemented a custom version of Myers' diff algorithm** to auto-detect writing syntax and generate personalized visual diffs of tabular data across three levels of granularity.

PROJECTS

Monkey Pose Estimation — Python, Pytorch, PyTorch Lightning, Hydra

- Trained state-of-the-art deep learning models for the pose estimation task, including HRNet and convolutional pose machines, on the OpenMonkeyChallenge dataset.
- Achieved a mean per joint position error (MJPE) score of **0.071**, probability of correct keypoint (PCK) score of **0.991**, and mean average precision (mAP) score of **0.786**.

SKILLS

Programming Languages: Python, Java, SQL, Matlab, HTML, CSS, JavaScript, TypeScript

Frameworks/Technologies: ReactJS, NodeJS, Flask, FastAPI, Pytorch, Pandas, Git, Latex

EDUCATION

University of Minnesota-Twin Cities

Aug 2024

Bachelor of Science in Computer Science and Mathematics

GPA: 3.8/4.0

AWARDS

1st Place Winner — Minnehack 2021 Hackathon

January 2021

Top Harmony Hack — LionHACK2022 Hackathon

April 2022

Ava Labs 3rd Place Winner — LionHACK2023 Hackathon

April 2023

AIME Qualifier

2018